TECHZIBITION

ABSTRACT

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ABSTRACT:

This project focuses on the development of a gesture-controlled car operated using hand movements. An accelerometer sensor detects the hand gestures and the ESP sends signals wirelessly through ESP NOW Protocol to a receiver connected to the second ESP microcontroller. The microcontroller processes these signals and controls the car's motors using a motor driver, allowing movement in different directions. This system provides a simple and intuitive way to control a vehicle without physical contact. The project demonstrates the use of sensors, wireless communication, and embedded systems, and has applications in robotics, automation, and assistive technologies.

How It Works:

- An accelerometer sensor detects the hand gestures by sensing the tilt and movement of the hand.

- The sensor sends this data to a microcontroller (ESP), which processes the signals.

- The processed signals are transmitted wirelessly ESP NOW.

- The receiver ESP on the car receives these signals and control the on the car.

- The microcontroller controls the motor driver module based on the received commands.

- The motor driver operates the motors, causing the car to move forward, backward, left, or right according to the hand gesture.

- This allows the car to be controlled wirelessly using simple hand movements without a physical remote.

COMPONENTS USED:

- ESP Microcontroller (ESP32) – Used to process signals from the accelerometer and control the car's movement.

- Accelerometer Sensor (MPU6050) – Detects hand gestures and tilt direction.

- Motor Driver Module (L298N) – Controls the speed and direction of the DC motors.

- DC Motors – Provide motion to the car wheels.

- Car Chassis with Wheels – Mechanical body of the car that holds all components.

- Battery (9V / Li-ion / Li-Po) – Provides power supply to the controller and motors.

- Glove or Handheld Controller – Used to mount the accelerometer sensor and ESP32 for gesture input.

- The Bot is in Progress some extra components may be used.

Why This Approach:

- Provides an intuitive way of controlling the car using simple hand gestures.

- Eliminates the need for traditional remote controllers and buttons.

- Demonstrates the practical use of sensors, microcontrollers, and wireless communication.

- Improves user interaction and makes the system more user-friendly.

- Can be useful in real-world applications such as robotics, assistive devices, and automation.

- Helps in understanding embedded systems and gesture recognition technology.

- Offers flexibility and wireless control, making the system efficient and modern.